Task 5: .NET Core Web API

Intern: Milan Tiwari | Organization: 366Pi Dev | Date: March 13, 2026 | Status: COMPLETED

1. Objective

Develop a **.NET Core Web API** to serve the rig data stored in the local MySQL database (synced in Task 4) to front-end applications via **JSON endpoints**.

MySQL Database (Data) → .NET Web API → JSON Response (Front-end)

2. Technology Stack

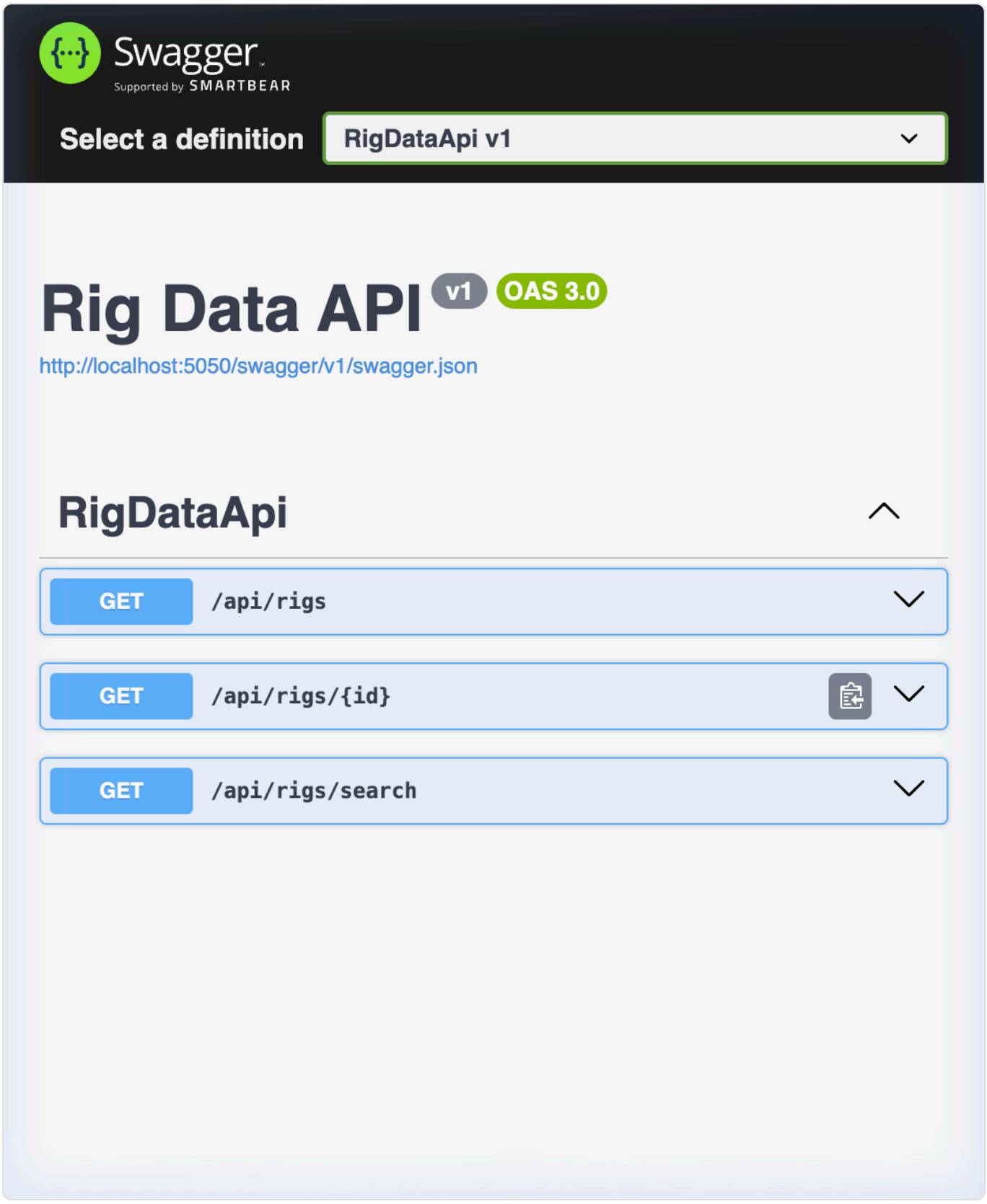
Technology	Purpose	Why This Choice
.NET 8 (C#)	API Framework	Minimal APIs for clean, high-performance code.
Dapper ORM	Database Access	Micro-ORM for fast, plain-SQL data retrieval.
MySql.Data	MySQL Connector	Official driver for MySQL connectivity.
Swagger/OpenAPI	Testing & Documentation	Interactive UI for discovering and testing endpoints.
CORS Policy	Security	Enabled to allow the React SPFx form to call the API.

3. Implementation Details

- **Scaffolded Project**: Created `RigDataApi` using the .NET 8 Web API template.
- **Data Model**: Defined `RigCompany` class matching the MySQL table schema.
- **Endpoints Created**:
  - `GET /api/rigs` — Returns all 9 companies as a JSON array.
  - `GET /api/rigs/{id}` — Returns a single company by its unique ID.
  - `GET /api/rigs/search?name=X` — Search companies by name.
- **Swagger UI**: Configured for automated documentation at `/swagger`.

4. Evidence of Success

Swagger UI Documentation



API JSON Response

Pretty print

[{"id":1,"companyName":"Noble Corporation plc","headquartersAddress":"3rd Floor, 1 Ashley Road, Altrincham, Cheshire, UK (WA14 2DT)","operationalRigs":37,"operationalJackUpRigs":12,"operationalMODUs":25,"officialWebsite":"noblecorp.com"},
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5. Challenges & Solutions

Challenge	Solution
DEPENDENCY	

Browser CORS Errors	Configured a **CORS Policy** in `Program.cs` to allow requests from the SPFx front-end.
VERSIONING Node.js Version Conflict	Discovered during manual run testing that Task 3 required Node v18 while system was on v24. Installed **NVM** to manage and swap versions.
ARCHITECTURE Fast Data Retrieval	Selected **Dapper** instead of Entity Framework to minimize overhead and ensure lightning-fast JSON responses.

6. Key Learnings

- **Minimal APIs** — Understanding how to build modern, lightweight microservices without MVC complexity.
- **JSON Serialization** — How .NET automatically converts C# objects into JSON for the browser.
- **Interactive Documentation** — Using Swagger to provide a user-friendly interface for API testers.
- **Unified Data Access** — Successfully bridging the gap between local database storage and front-end consumption.